Event Streams of Moving Scenes: Rate Limits and Regeneration Under Affine Motion

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Abstract

An event camera reports the changes of each pixel's log intensity as asynchronous threshold crossings. When a scene moves across the sensor, every pixel on the path of an edge observes the same intensity profile, so the stream appears to describe that profile many times. It does not repeat exactly: each pixel fires on its own lattice of threshold levels, whose offset, the threshold phase, differs from pixel to pixel, and every event carries timing jitter. We determine what these differences cost a code, for a ramp edge under an affine flow. Losslessly, the phase of each pixel and the jitter of its other events must be described however well the scene and its motion are known, so the rate per event does not fall with the length of the path. Under a mean-squared timing tolerance there are three regimes. Below the jitter variance the code describes every event time. Above it the code describes one number per pixel. Beyond a threshold set by the event spacing of the pixel it describes nothing, because the events can be regenerated from the edge and its motion, and what remains is the motion, at a cost that grows at most logarithmically with the path. The rate-distortion function is exact in the first regime and known within 0.255 bit per pixel in the second. These results lead to REMAKE, a codec that transmits the events of one window as a template and one affine map for each later window, and we quantify what moving events loses against regenerating them. On recordings from a static camera and from a moving car, with regions that the sender finds itself and with the bytes of a complete stream as the cost, REMAKE needs fewer bits than schemes that send a subset of the events at equal fidelity once a group holds four windows or more, and its advantage shrinks as the time bin of the receiver shortens.

Index Terms: Event camera, send-on-delta sampling, level-crossing sampling, rate-distortion theory, generalized sampling, motion compensation, neuromorphic sensing.

1 Introduction

An event camera has no frames. Each pixel compares its log intensity with a stored reference and emits an event, with a time stamp and a polarity, whenever the difference reaches a contrast threshold [1,2]. The output is a stream of address-events whose rate follows the scene: it is close to zero for a static scene and rises with the speed and the contrast of what moves. A sensor node that forwards this stream over a rate-limited link must decide how many bits the stream needs.

The stream of a moving scene looks highly redundant. If an object moves across the sensor, the pixels along the path of one of its edges observe the same intensity profile, one after the other. Yet they do not emit the same events. Each pixel compares the intensity with its own reference, so it fires on its own lattice of levels, and the offset of that lattice, the threshold phase of the pixel, differs from pixel to pixel. The same profile therefore produces a different event train at every pixel (Fig. 1). This paper asks what these differences cost: how many bits a code needs to preserve them, and how many it saves when a timing error is allowed.

The setting has precedents in sampling theory. An event pixel is a send-on-delta sampler [3], which for a monotone input coincides with level-crossing sampling [4, 5]. An object that crosses M pixels is sampled by M such samplers, which recalls the generalized sampling expansion [6], by which a bandlimited signal is recovered from the outputs of m linear systems that are each sampled at $1/m$ of the Nyquist rate. In multi-channel time encoding, several integrate-and-fire encoders with shifted integrators share one input, and the shifts are what make the channels complementary [7–9]. The threshold phases play that role here.

Existing event coders approach the rate empirically. Lossless coders exploit the spatial and temporal correlation of addresses and time stamps [10–12]. Lossy coders discard or aggregate events [13, 14], and the coder of [15] suppresses events that an on-sensor optical flow can predict and lets the receiver extrapolate them. A recent survey notes that reported results are hard to compare [16], and no accepted distortion measure for lossy event coding exists [17]. On the theoretical side, rate-distortion results for point processes concern Poisson and finite point processes [18–21]. They do not model events that a moving profile generates through a threshold rule.

This paper models that mechanism and derives what a code can and cannot save. Its central result is the rate-

distortion function of the events of a pixel under a timing tolerance. The other results lead to it or follow from it.

1) Mechanism. Given the edge and its motion, the events of a pixel are a function of a single number, its threshold phase, up to timing jitter. The phase is a property of the pixel and is independent across pixels. Next to what the pixels of a path share, the edge and its motion, each pixel therefore has something private. A lossless code must describe it for every pixel, and its rate per event does not decrease with the number M of pixels on the path (Theorem 1).

2) Rate under a timing tolerance. The tolerance decides which of the private quantities a code must describe (Theorem 2). Below the jitter variance it describes all of them, and the rate is known exactly. Above it one number per pixel remains, and the rate is known within 0.255 bit. Beyond a threshold that grows with the event spacing of the pixel, the contrast threshold divided by the rate of change of its intensity, the rate is zero: the decoder regenerates the events from the edge and its motion. Slowly crossed pixels need a larger tolerance. Along a path, a description of the motion of $O(\ln M)$ nats then replaces a lossless description of order M.

3) Regeneration in practice. A sender has the events and not the edge. We replace the edge by the events of a key window, which the receiver moves along a fitted affine map. Moved events keep the phases of the pixels they came from, and we show that this doubles the timing error of the ideal decoder at equal event spacings (Proposition 2). The sender finds its regions on a fixed block grid, and the stream is complete, so that its bytes are the cost. We measure this codec, REMAKE, on recordings from a static camera [22] and from a moving car [23], against schemes that send a subset of the events.

Section 2 states the model. Sections 3 and 4 contain the theory. Section 5 describes REMAKE, and Section 6 the measurements. Proofs are in the appendices. Rates are in nats unless bits are named, $\|\cdot\|$ is the Euclidean norm of a vector and the spectral norm of a matrix, and $[x]^+ = \max(x, 0)$.

2 Model

We first state the pixel rule, from which the phase mechanism follows for any scene. We then specialize to a ramp edge in uniform passage, for which the rates have closed forms.

2.1 Pixel

Pixels sit at the lattice points $i \in \mathbb{Z}^2$ of the image plane. Pixel i observes the log intensity $\ell_i(t) = L(i, t)$ of a continuous field L.

(A1) Send-on-delta. Pixel i holds a reference μ_i with $|\ell_i - \mu_i| < C$ at the start. It emits an event at the first time t at which $|\ell_i(t) - \mu_i| = C$, with polarity $p = \operatorname{sign}(\ell_i(t) - \mu_i)$, and then sets $\mu_i \leftarrow \mu_i + pC$. The contrast threshold $C > 0$ is common to all pixels and known to the decoder. The *phase* of the pixel is the number $a_i \in (0, C]$ with $\mu_i \in a_i + C\mathbb{Z}$ at the start.

2.2 Scene and Motion

A template $g : \mathbb{R}^2 \to \mathbb{R}$ gives the log intensity in object coordinates. At time t the object point u is at the image position $\phi_t(u) = J_t u + b_t$, with $J_t \in \mathbb{R}^{2\times 2}$ invertible, so

$$L(x, t) = g\big(\phi_t^{-1}(x)\big) \tag{1}$$

and pixel i observes

$$\ell_i(t) = g\big(u_i(t)\big), \qquad u_i(t) = J_t^{-1}(i - b_t). \tag{2}$$

The curve u_i is the trajectory of pixel i in object coordinates. The velocity field $v(x, t) = \dot{\phi}_t(\phi_t^{-1}(x))$ is affine in x and satisfies the brightness constancy relation

$$\frac{\partial L}{\partial t}(x, t) + \langle \nabla L(x, t), v(x, t)\rangle = 0, \tag{3}$$

which follows from differentiating $L(\phi_t(u), t) = g(u)$ in t. We call the motion an *affine flow* when the velocity field is also constant in time,

$$v(x) = \Lambda x + v_0, \tag{4}$$

with generator $\Lambda \in \mathbb{R}^{2\times 2}$. A translation has $\Lambda = 0$. The trace of Λ is the rate at which the object grows or shrinks in area, its antisymmetric part rotates the object, and its symmetric traceless part shears it. At a given instant, any smooth velocity field has the form (4) to first order in x around a point, so (4) is the local model of a general motion over a short time.

2.3 Edge

(A2) Ramp edge. The template is a ramp across a line with unit normal ν,

$$g(u) = \varrho\big(\langle \nu, u\rangle - c_0\big), \qquad \varrho(z) = \begin{cases} 0, & z \le 0, \\ \gamma z, & 0 \le z \le w, \\ V, & z \ge w, \end{cases} \tag{5}$$

with slope $\gamma > 0$, width w and height $V = \gamma w = nC$ for an integer $n \ge 1$.

Pixel i sees the edge through its edge coordinate

$$z_i(t) = \langle \nu, u_i(t)\rangle - c_0, \qquad \ell_i(t) = \varrho\big(z_i(t)\big). \tag{6}$$

(A2′) Uniform single passage. The edge passes pixel i once: $z_i < 0$ before a time t_i, $z_i > w$ after the passage, and in between z_i is affine in t with slope $r_i > 0$.

Inside the ramp, $\nabla L = \gamma \nu_t$ with $\nu_t = J_t^{-\top}\nu$ a normal of the edge in the image, and $\dot{z}_i = -\langle \nu_t, v(i, t)\rangle$. By (3), the intensity of pixel i rises during the passage at the rate

$$\gamma r_i = -\langle \nabla L, v\rangle = \|\nabla L\|\, v_{\perp,i}, \tag{7}$$

with the field evaluated at pixel i. Here $v_{\perp,i} = -\langle \nabla L, v\rangle / \|\nabla L\|$ is the speed at which the edge advances over the pixel along its normal, counted positive when the intensity of the pixel rises, as it does in a passage with $r_i > 0$. We call

$$\tau_i = \frac{C}{\gamma r_i} = \frac{C}{\|\nabla L\|\, v_{\perp,i}} \tag{8}$$

the *event spacing* of pixel i. It is the time that the ramp takes to pass the pixel, divided by n. At a fixed image slope $\|\nabla L\|$ it is inversely proportional to the normal speed of the edge. Assumption (A2′) holds exactly for a translation, where τ_i is the same for all pixels, and for a plane that approaches the camera along the optical axis at constant speed, where J_t^{-1} is affine in t and b_t is fixed. For an affine flow it holds to first order (Remark 1), and τ_i varies from pixel to pixel.

2.4 Phase, Jitter and Clock

(A3) Phase. Before the edge arrives, pixel i rests at the level 0 with its reference behind the passage, $\mu_i \in (-C, 0]$, so that $a_i = \mu_i + C$. The phases a_i are independent across pixels, uniform on $(0, C]$, and independent of the jitter.

(A4) Jitter and clock. The kth event of pixel i is recorded at $T_{i,k} = t_{i,k} + \varepsilon_{i,k}$, where $t_{i,k}$ is the time given by (A1) and the $\varepsilon_{i,k}$ are independent $\mathcal{N}(0, \sigma^2)$ variables, independent of the phases, with $\sigma > 0$. The stored value is $Y_{i,k} = \delta \lfloor T_{i,k}/\delta \rfloor$ for a clock resolution δ. The index k of each recorded event is known.

The uniform phase does not arise inside the model, in which every pixel rests at the same level. It stands for the offsets of real pixels and for their earlier activity. A pixel that the edge crosses from the high level to the low level is covered by the mirrored assumption, with its reference in $[V, V + C)$. Remark 4 treats a reference that is ahead of the passage.

We write $T_i = (T_{i,1}, \dots, T_{i,n})$ and $Y_i = (Y_{i,1}, \dots, Y_{i,n})$, and $\mathcal{P}$ for the set of pixels that the edge passes during the observation, with $M = |\mathcal{P}|$. Let $\eta(a, \sigma)$ denote the differential entropy of the sum of a uniform variable on $(0, a)$ and an independent $\mathcal{N}(0, \sigma^2)$ variable. It satisfies

$$\tfrac{1}{2} \ln\!\left(a^2 + 2\pi e \sigma^2\right) \le \eta(a, \sigma) \le \tfrac{1}{2} \ln\!\left(2\pi e (a^2/12 + \sigma^2)\right) \tag{9}$$

by the entropy power inequality [24, Ch. 17] and by the Gaussian bound on the entropy at a given variance [24, Ch. 8], and $\eta(a, \sigma) - \ln a \to 0$ as $\sigma/a \to 0$.

3 Structure and Lossless Limit

Lemma 1. *Under (A1), for any continuous ℓ_i, every event of pixel i occurs at a time t with $\ell_i(t) \in a_i + C\mathbb{Z}$.*

The lattice $a_i + C\mathbb{Z}$ never changes. The phase is a property of the pixel, set by its history and not by the edge that passes.

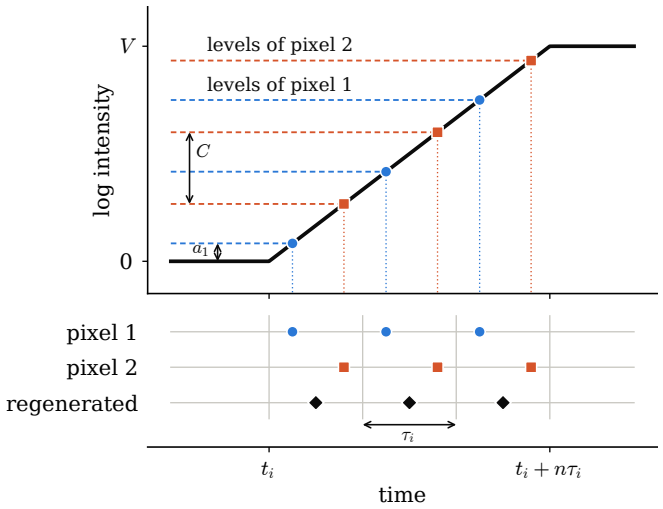


Figure 1: One ramp, two pixels with different threshold phases, and the regenerated events (22). Time is measured from the arrival of the edge at each pixel. Jitter is omitted.

Lemma 2. *Under (A1), (A2), (A2′) and $\mu_i \in (-C, 0]$ before the passage, pixel i emits exactly n events, all of positive polarity, at*

$$t_{i,k} = t_i + A_i + (k-1)\tau_i, \qquad k = 1, \dots, n, \tag{10}$$

where $A_i = a_i/(\gamma r_i)$. Under (A3), A_i is uniform on $(0, \tau_i]$.

Given the template and the flow, the noise-free events of a pixel are thus a function of one number, its phase. Fig. 1 shows two pixels with $n = 3$ that see the same ramp: the intensity trajectories coincide, and the event trains differ by the difference of the phases. In the object coordinate, an event is a point of the graph of the profile on the level lattice of its pixel: the M pixels of a path sample the same profile on M lattices with independent offsets.

Theorem 1 (Lossless code length)**.** *Under (A1) to (A4) and (A2′), the vectors T_i, $i \in \mathcal{P}$, are independent given the template and the flow,*

$$h(T_i) = \frac{n-1}{2} \ln(2\pi e \sigma^2) + \eta(\sqrt{n}\,\tau_i, \sigma), \tag{11}$$

and

$$H(Y_i \mid g, \phi) = n \ln \frac{1}{\delta} + h(T_i) + e_\delta, \tag{12}$$

where $e_\delta \ge 0$ for every δ, and $e_\delta \to 0$ as $\delta/\sigma \to 0$, uniformly in τ_i.

When $\sigma \ll \tau_i$ and $\delta \ll \sigma$, (12) gives the cost per event

$$\frac{H(Y_i \mid g, \phi)}{n} \approx \frac{n-1}{n} \ln \frac{\sigma\sqrt{2\pi e}}{\delta} + \frac{1}{n} \ln \frac{\sqrt{n}\,\tau_i}{\delta}. \tag{13}$$

One event of each pixel pays for the phase and the other $n-1$ pay for jitter. The code length of the path is the sum

of (12) over $\mathcal{P}$. It grows in proportion to M, and neither term of (13) falls as the path gets longer: a decoder that knows the edge and its motion exactly still pays (13) for every event. What the pixels of a path share, the edge and its motion, a code can describe once. What is private to each pixel, its phase and its jitter, it must describe for each pixel, and (12) is that irreducible conditional rate. Section 6 measures how much more a code pays that does not know the motion.

Remark 1 (Accuracy of the uniform passage). For an affine flow (4), $\dot{\nu}_t = -\Lambda^\top \nu_t$ and $\ddot{z}_i = \langle \nu_t, \Lambda v(i) \rangle$, hence $|\ddot{z}_i| \le \kappa_i |\dot{z}_i|$ with $\kappa_i = \|\Lambda\|/|\cos\vartheta_i|$, where ϑ_i is the angle between the velocity $v(i)$ and the normal ν_t. Take r_i as the slope of z_i at t_i. Over a passage of duration $P_i = n\tau_i$, the event times then deviate from (10) by at most $\kappa_i P_i^2/2$, to first order in $\kappa_i P_i$. For a generator of norm $0.3\,\mathrm{s}^{-1}$, a passage of 5 ms and motion along the normal this is $4\,\mu$s, three orders of magnitude below the passage time. The bound grows without limit for edges that are nearly parallel to their motion. Such edges pass few pixels per unit time and contribute few events.

Remark 2 (Several edges). Theorem 1 counts one edge that passes each pixel once. If further edges of the same object pass the same pixel, they meet the same lattice, so the phase is paid once per pixel and not once per edge. With many edges per pixel the cost per event tends to the jitter term $\ln(\sigma\sqrt{2\pi e}/\delta)$, which does not depend on the path either.

4 Rate Under a Timing Tolerance

The encoder and the decoder know the template and the flow. The decoder outputs $\hat{T}_i \in \mathbb{R}^n$ for each pixel, and the distortion is the mean-squared timing error

$$d(T_i, \hat{T}_i) = \frac{1}{n}\|T_i - \hat{T}_i\|^2. \tag{14}$$

4.1 One Pixel

Let $R_i(D) = \inf I(T_i; \hat{T}_i)$, with the infimum over the conditional laws of $\hat{T}_i$ that satisfy $\mathbb{E}\, d(T_i, \hat{T}_i) \le D$, be the rate-distortion function of the events of pixel i. It is the rate that block codes over many independent passages approach. Define

$$s_i^2 = \frac{n\tau_i^2}{12} + \sigma^2, \qquad D_{A,i} = \sigma^2 + \frac{\tau_i^2}{12}, \tag{15}$$

$$\Gamma_i = \tfrac{1}{2}\ln(2\pi e\, s_i^2) - \eta(\sqrt{n}\,\tau_i, \sigma). \tag{16}$$

By (9), $0 \le \Gamma_i \le \frac{1}{2}\ln(2\pi e/12)$, which is 0.255 bit, for every σ and τ_i.

Theorem 2 (Rate-distortion function of a pixel). *Under (A1) to (A4) and (A2′),*

$$R_i(D) = h(T_i) - \frac{n}{2}\ln(2\pi e D) \qquad \text{for } 0 < D \le \sigma^2, \tag{17}$$

$$\big[U_i(D) - \Gamma_i\big]^+ \le R_i(D) \le U_i(D) \qquad \text{for } \sigma^2 < D < D_{A,i}, \tag{18}$$

$$R_i(D) = 0 \qquad \text{for } D \ge D_{A,i}, \tag{19}$$

with

$$U_i(D) = \frac{1}{2}\ln\frac{s_i^2}{nD - (n-1)\sigma^2}. \tag{20}$$

Moreover,

$$R_i(D) \ge \frac{n\,(D_{A,i} - D)}{2s_i^2} \qquad \text{for } 0 < D \le D_{A,i}, \tag{21}$$

so that $R_i(D) > 0$ for $D < D_{A,i}$.

The three regimes have a direct reading. Below the jitter variance, every bit of timing precision costs a bit, per event: the rate falls by $\log_2 10 = 3.32$ bits per event and per decade of the root-mean-square tolerance $\sqrt{D}$. Between σ^2 and $D_{A,i}$, the $n-1$ jitter components orthogonal to a common shift of the events are no longer described, and the code describes one number per pixel, the phase together with the mean jitter, to within 0.255 bit. From $D_{A,i}$ on, nothing is described. The decoder that attains (19) outputs the mean times

$$\hat{T}_{i,k} = t_i + \big(k - \tfrac{1}{2}\big)\tau_i, \qquad k = 1, \dots, n. \tag{22}$$

It *regenerates* the events of the pixel from the edge and the flow. Each regenerated event is within $\tau_i/2$ of its noise-free time, and the count and the polarities are exact.

Two of these statements extend beyond the ramp. If the noise-free event times of a pixel are any function of its phase, (17) holds below the jitter variance with the entropy of the actual event times, and the smallest distortion at zero rate is $\frac{1}{n}\operatorname{tr}\operatorname{Cov}(T_i)$, attained by the mean times. The middle regime, in which one number per pixel is described to within 0.255 bit, and the closed forms (11), (20) and (15) belong to the ramp in uniform passage, where the phase shifts all events of a pixel by a common time.

The threshold $D_{A,i}$ grows with the square of the event spacing. By (8), a pixel whose intensity changes slowly has a large spacing and needs a large tolerance before its events can be regenerated, and a pixel that is crossed quickly needs a small one.

4.2 A Path

For the M pixels of a path, let the distortion be the average of (14) over $i \in \mathcal{P}$, and let $R_{\mathcal{P}}(D)$ be the rate-distortion function of $(T_i)_{i\in\mathcal{P}}$ under it.

Theorem 3 (Rate-distortion function of a path). *Under (A1) to (A4) and (A2′),*

$$R_{\mathcal{P}}(D) = \min\Big\{ \sum_{i\in\mathcal{P}} R_i(D_i) : \ \frac{1}{M}\sum_{i\in\mathcal{P}} D_i \le D \Big\}. \tag{23}$$

In particular,

$$R_{\mathcal{P}}(D) = \sum_{i\in\mathcal{P}} h(T_i) - \frac{Mn}{2}\ln(2\pi e D) \qquad \text{for } 0 < D \le \sigma^2, \tag{24}$$

and $R_{\mathcal{P}}(D) = 0$ *if and only if*

$$D \;\ge\; \bar{D}_A = \sigma^2 + \frac{1}{12M}\sum_{i\in\mathcal{P}} \tau_i^2. \tag{25}$$

If $\tau_i \le \tau_{\max}$ *on the path, then for* $D \le \bar{D}_A$

$$R_{\mathcal{P}}(D) \ge \frac{nM\,(\bar{D}_A - D)}{2\,(n\tau_{\max}^2/12 + \sigma^2)}. \tag{26}$$

If the tolerance is imposed on every pixel separately, the rate is zero if and only if $D \ge \max_i D_{A,i}$, and the slowest crossing sets the tolerance. By (26), below $\bar{D}_A$ a rate that is proportional to M remains.

4.3 The Motion

Theorems 2 and 3 assume that the decoder knows the flow. It remains to count what describing the flow costs.

Proposition 1 (Description of the motion)**.** *Let the flow depend on a parameter* $\theta = (\theta_1, \dots, \theta_q)$ *that ranges over a box whose* j*th side has length at most* $L_j M^{\alpha_j}$*, and let the mean times* $\bar{t}_{i,k}(\theta) = t_i(\theta) + (k - \frac{1}{2})\tau_i(\theta)$ *of* (22) *satisfy* $|\partial \bar{t}_{i,k}/\partial\theta_j| \le S_j M^{\beta_j}$ *on the box, for all events of the path, with constants* $L_j, S_j > 0$ *and exponents* $\alpha_j, \beta_j \ge 0$ *that do not depend on* M*. Then for every* $\xi > 0$ *the events of the path can be described with average distortion at most* $(1+\xi)\bar{D}_A$ *using*

$$B \le \Big(\sum_{j=1}^{q} (\alpha_j + \beta_j)\Big)\ln M + O(1) \quad \textit{nats}, \tag{27}$$

where the $O(1)$ *term depends on* q*,* ξ*,* σ *and the constants and not on* M*.*

For a translation along a line with a speed in a fixed interval that excludes zero, θ holds the position of the edge, with a range of order M and a sensitivity of order one, and the speed, with a range of order one and a sensitivity of order M, so $B \le 2\ln M + O(1)$. The same holds with finite exponents, and hence $B = O(\ln M)$, for an affine flow whose normal speed is bounded away from zero on the path and whose duration is polynomial in M. The lossless cost of the same events is the sum of (12), of order M, so the ratio of the two grows at least as $M/\ln M$. Above $\bar{D}_A$ the description that remains is $O(\ln M)$. Below it, a rate proportional to M remains. The bound (27) is for a receiver that holds the edge. Section 5 accounts for the template and the regions.

Remark 3 (Event count under a change of scale). An edge segment emits n events for every pixel that it passes, so its event rate is n times the area that it sweeps per unit time, that is n times its length times its mean normal speed. Under a similarity flow about a fixed point, growth by a factor ζ makes the segment ζ times longer and ζ times faster, and the rate grows as ζ^2. For an object that approaches at constant speed, the image speed itself grows as ζ^2 and the rate as ζ^3. A regeneration that moves a fixed set of events keeps their number. Under a change of scale it must also change the number of events, or regenerate them from the edge.

Remark 4 (Departures from the model). *Reference ahead of the passage.* If $\mu_i \in (0, C)$ before a rising passage, as after a passage of the opposite polarity at the same pixel, the first level that the intensity meets is the reference itself, which does not fire. The pixel then emits $n-1$ events, and the results hold with $n-1$ in place of n and $t_i + \tau_i$ in place of t_i. *Height.* If the height of the edge is not a multiple of the threshold, $V/C = n + f$ with $0 < f < 1$, pixel i emits $n+1$ events when $a_i \le fC$ and n otherwise. Reproducing the count exactly costs the binary entropy of f per pixel, and a zero-rate statement then holds only for a distortion measure that tolerates one unmatched event per pixel, with a modified threshold. *Sensor.* Noise events, threshold mismatch between pixels, a refractory time and a jitter that depends on the slope are outside the model, and their effect on the three regimes is not analyzed here.

5 A Regenerating Codec

The decoder (22) needs the edge and the flow. A practical sender has neither. It has the events. The codec of this section, REMAKE (REgeneration by Mapping Affine Key Events), replaces the edge by events that the receiver already holds and the flow by maps that the sender fits, and it finds its regions without any annotation.

5.1 Regions, Key Events and Groups

Time is cut into windows of length T_0, and consecutive windows form groups of K. The sensor is cut into square blocks of $B \times B$ pixels. In the first window of a group, the key window, the blocks that hold at least ψB^2 events are the regions of the group, where the threshold ψ is a density in events per pixel. The sender transmits the events of the regions, the key events, and for each of the following $K-1$ windows one affine map per region. The receiver moves every key event (x, p, t) of a region to the pixel nearest to the image of x under the map of that region and to the time $t + mT_0$, where m counts the windows since the key window, and it keeps the polarity. A moved event may leave its block. One that leaves the sensor is dropped. The moved events are the reconstruction of window m. The regions stay fixed within a group, so what enters the scene after the key window is not reconstructed before the next one.

Under an affine flow, the map from one instant to an

instant mT_0 later is the same for all instants of a window, so one map per window is exact for the positions of the edges of a region that moves with the flow. It does not change the number of events, which a change of scale does (Remark 3).

5.2 Maps and Their Fit

With c the center of the block, a map moves the position x to

$$x' = x + d + \frac{2}{B} A\,(x - c), \tag{28}$$

where $d \in \mathbb{Z}^2$ is a displacement in pixels and $A \in \mathbb{Z}^{2\times 2}$ holds four shape parameters. Each of them is a displacement in pixels at the edge of the block: the diagonal of A stretches the block along the two axes, and the other two entries shear or rotate it. With $A = 0$ the map is a translation.

The sender fits the maps of a window region by region, by a greedy search for the number of moved events that can be paired with true events of that window inside a voxel of one fixed working grid, which is the numerator of the fidelity (30). It visits the regions in the order of decreasing event count, and a true event that an earlier region has paired is not available to a later one, so the pairs of the regions add up to the pairs of the whole reconstruction. For each region a search from coarse to fine finds the displacement, from zero in the first window and from the extrapolated map of the previous window afterwards, and a coordinate search then finds the shape parameters. On the true events that are still unpaired when a region is visited, its map never pairs fewer events than the translation it started from, and that translation never fewer than no motion. Results on other grids use the same maps.

5.3 Bitstream and Event Coders

A stream is a header with the configuration (the sensor size, B, K, T_0, the kind of map and the event coder), followed by one record per group: the index of its key window, the key events and the maps, each of the two preceded by its length. The regions are not listed. They are the blocks that hold key events, and the maps follow in the raster order of these blocks, as differences along m in a variable-length integer code, compressed with Zstandard [25]. The bits of a group are eight times the bytes of its record. The header, 28 bytes per stream, is less than 10^{-3} of a stream and is left out. A record is complete when the last window of its group has arrived, so the sender delays a stream by one group. Every fidelity of REMAKE in Section 6 is measured on events that a decoder rebuilds from the bytes alone.

Two event coders serve the key events and all schemes that we compare with. The *transport coder* keeps time stamps of one microsecond: it packs the addresses, writes the time differences in a variable-length code and applies Zstandard at its first level. The *matched coder* is for a receiver that counts events on a known grid: it sorts the events of a window by voxel and writes the gaps between voxel indices in a variable-length code, followed by Zstandard at level 19. A subset of the events is then coded on the grid of the receiver. The key events are coded at one pixel and the time bin of that grid, because a map moves them by whole pixels and keeps their time bin.

5.4 Moved Events Against Regenerated Events

REMAKE differs from the decoder (22) in three ways. It moves the phases of the key pixels where (22) outputs the mean over the phase, it moves the jitter of the key events, and it moves a fixed number of events. The first two have a price that the model gives exactly.

Proposition 2 (Moved events)**.** *Let the same edge pass pixel j in the key window and a different pixel i later, under (A1) to (A4) and (A2′), and let the map be exact, so that the kth event of pixel j is moved to pixel i at the time $\tilde{T}_{i,k} = T_{j,k} + t_i - t_j$. Then*

$$\frac{1}{n}\,\mathbb{E}\|T_i - \tilde{T}_i\|^2 = D_{A,i} + D_{A,j} + \frac{4n^2 - 1}{12}\,(\tau_i - \tau_j)^2. \tag{29}$$

With equal event spacings the error is $2D_A$, twice that of (22): REMAKE trades a factor of two in the timing error for not needing the edge. Unequal spacings, as under a change of scale, add the last term, which grows with the square of the number of events per pixel. If the moved events fall on their own pixel, as for an object that is nearly at rest, the phases coincide and only the jitter term $2\sigma^2$ remains. The third difference, the number of events, is not a timing error. It is measured in Section 6.

5.5 Receiver Representation, Fidelity and Baselines

The theory uses the timing error (14), which presupposes that regenerated and true events are paired. For measured streams we use the representation that event-based receivers compute, a voxel grid: the number of events of each polarity in blocks of $\Delta_x \times \Delta_x$ pixels and in time bins of length Δ_t. For a reconstruction with voxel counts c' and a true set with counts c, of sizes N' and N, the fidelity is

$$F = \frac{2\sum \min(c', c)}{N' + N}, \tag{30}$$

the share of the events of the two sets that can be paired one to one inside a voxel. It is computed over the complete sensor, so the events outside the regions count as lost. A subset that holds a share χ of the events has $F = 2\chi/(1 + \chi)$ on every grid, whichever events it keeps. A reconstruction of fidelity F is therefore matched by a subset of the share

$$\chi_F = \frac{F}{2 - F}. \tag{31}$$

Schemes that send a subset of the events and regenerate nothing are the baseline. We measure three: a random subset, complete windows with the other windows dropped, and, in every window, the blocks that the rule of the key window selects. Two such schemes can be mixed over windows, which mixes their shares and their bits linearly, so the cheapest subset scheme at a share χ costs $b_{\mathrm{sub}}(\chi)$ bits per event, the lower convex envelope of the measured operating points and the origin. For a codec that spends b bits per event of a group and reaches the fidelity F over the group, the key window included, we report

$$G = \frac{b_{\mathrm{sub}}(\chi_F)}{b}, \tag{32}$$

the factor by which it lowers the bits at equal fidelity. A random subset alone is a weaker baseline than the envelope: its events lie further apart than those of whole windows or whole blocks, and each of them costs more bits.

The fidelity (30) and the timing error (14) are different distortion measures, and the bin length plays the role of the tolerance. One case connects them. For blocks of one pixel, a train that is long against the event spacing, and a jitter that is small against the bin, Lemma 2 gives the fidelity of the regenerated events (22) directly: a bin holds at most one event of each train when $\Delta_t \le \tau_i$, the offset between the two trains is uniform, and in expectation over the phase

$$F = \frac{\Delta_t}{\tau_i} \qquad \text{for } \Delta_t \le \tau_i, \tag{33}$$

with $F = 1$ at $\Delta_t = \tau_i$. The voxel counts of a pixel are thus reproduced only when the bin is at least as long as its event spacing, the same scaling as the threshold of Theorem 2, for which $\sqrt{D_{A,i}} = \tau_i/\sqrt{12}$ when $\sigma \ll \tau_i$. In larger blocks the independent phases of the pixels average, and the fidelity is higher at a given bin length.

6 Experiments

Section 6.2 checks Theorem 2 on the model. The other sections measure REMAKE on recorded streams.

6.1 Protocol

Data. The recordings from a static camera are the 112 recordings of the training split of eTraM [22], 61 by day and 51 by night, from a sensor of 1280×720 pixels above a road, with 2.4×10^{10} events. The recordings from a moving camera are the 41 training sequences of DSEC [23], left event camera, 640×480 pixels, on a car, with 3.7×10^{10} events in 2634 s. A window of the median DSEC sequence holds 1.15 events per pixel, against 0.04 for eTraM. Each recording is replayed as a stream, in the order of its events.

Codec. Windows have $T_0 = 33.3$ ms. Groups of $K = 2$, 4, 8 and 11 windows share their key window, which is taken every 3 s of an eTraM recording and every 2 s of a DSEC sequence. The working grid has $\Delta_x = 2$ pixels and $\Delta_t = T_0/3 = 11.1$ ms, and the event coder is the matched coder on that grid unless the transport coder is named. Each dataset is calibrated on a tuning subset of its own: twelve eTraM recordings, every tenth by day and by night, and five DSEC sequences, every tenth of the sorted list. On it the block side is selected among 20, 40 and 80 pixels and the threshold among 1/16, 1/8, 1/4, 1/2 and 1 event per pixel. The criterion is the bits that REMAKE saves against the subset schemes at equal fidelity for $K = 4$, as a share of the bits of all events. It selects $B = 20$ and $\psi = 1$ on both datasets. On DSEC the runner-up, $B = 40$ and $\psi = 1$, saves 0.086 of the bits of all events against 0.090. The configuration is then fixed, and every result below is over the other 100 eTraM recordings and 36 DSEC sequences. The selected threshold is the largest candidate. The saving grows with the threshold over the whole range while the fidelity falls, so the threshold acts as a rate control, and Section 6.3 reports the two ends of the range. Only B and ψ are calibrated, and T_0, K, the grids and the coders are set in advance. The regions, the key events and the maps are computed from the stream, for every group anew.

Statistics. A recording enters with the sums over its groups, and we report the median over recordings, with a 95% interval from a bootstrap over recordings where one is given. Of the 100 eTraM recordings, 93 hold the ten groups that we require (53 by day and 40 by night, 4958 groups). The DSEC sequences are shorter, so we require five groups there, and all 36 enter, with 1189 groups.

6.2 The Three Regimes

Fig. 2 compares Theorem 2 with a rate-distortion function that is computed without it, for one pixel with $n = 3$ and $\sigma/\tau_i = 0.05$. These values illustrate the theorem and are not those of a particular sensor. By (34), the event times are an orthogonal image of the scalar X_i, which carries the phase, and of $n - 1$ normal variables. The iteration of [26] gives the rate-distortion function of X_i on a grid of 1217 points, and the normal components enter in closed form at the same slope.

Below σ^2 the numerical rate equals (17) to within 4×10^{-6} nat. Between σ^2 and $D_{A,i}$ it stays inside the band between the bounds (18) and (21), which is at most $\Gamma_i = 0.187$ bit per pixel wide here, against the limit of 0.255 bit. The rate leaves σ^2 on the lower bound $U_i - \Gamma_i$: the phase is still described finely there, and the code profits fully from its distribution, which is uniform and not normal. Toward $D_{A,i}$ the description becomes coarse, the rate depends on the phase only through its variance, and the numerical rate, the bound (21) and U_i meet. From $D_{A,i}$ on the rate is zero, and the regenerated events (22) of 2×10^6 simulated pixels have the distortion $0.9994\, D_{A,i}$.

The squares are a uniform scalar quantizer with entropy coding, applied to what each regime describes. On

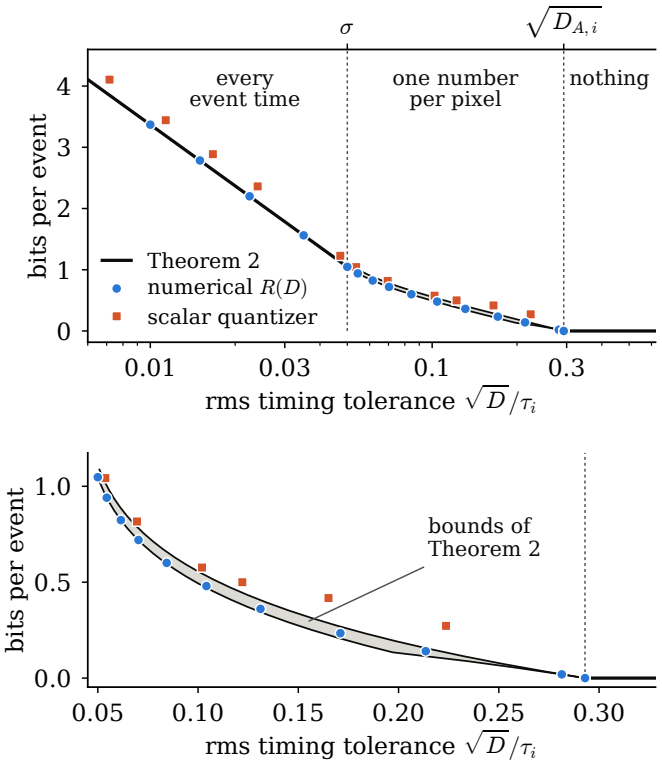


Figure 2: Rate of one pixel against the timing tolerance for $n = 3$ and $\sigma/\tau_i = 0.05$: Theorem 2, a numerical rate-distortion function and a scalar quantizer. Bottom: the regime between σ and $\sqrt{D_{A,i}}$.

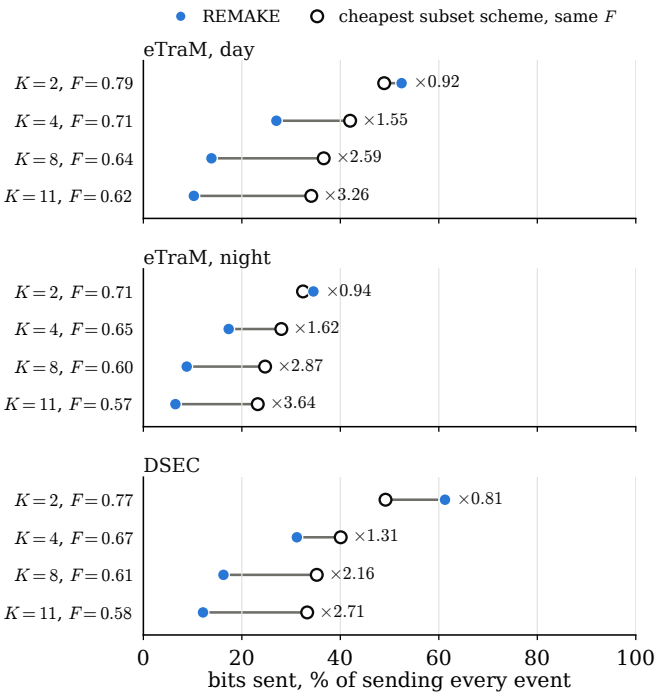


Figure 3: Bits of REMAKE and of the cheapest subset scheme at the fidelity F of REMAKE, as a percentage of the bits of all events, for groups of K windows. Matched coder, working grid. The labels give G.

every component of the event times it stays 0.254 to 0.258 bit per event above (17). This is the loss $\frac{1}{2}\ln(2\pi e/12)$ of a uniform quantizer at fine resolution, the difference between the entropy $h - \ln\Delta$ of the index of a quantizer of step Δ and (17) at the distortion $\Delta^2/12$. On the mean event time alone, with the other components left out, it stays 0.09 to 0.19 bit per event above the lower bound up to $D_{A,i}$. A code that describes every event time, then one number per pixel, then nothing, is therefore within 0.26 bit per event of the limit in all three regimes.

6.3 Total Bits Against Fidelity

Fig. 3 shows the bits of REMAKE against those of the subset schemes at its fidelity, and Table 1 gives the groups of four and of eleven windows.

Gain. On eTraM a group of four windows costs 0.42 bit per event by day and 0.35 by night and reaches the fidelities 0.71 and 0.65. The cheapest subset scheme needs 0.65 and 0.57 bit at these fidelities, so $G = 1.55$ (95% interval 1.52 to 1.57) by day and 1.62 (1.61 to 1.65) by night. The gain grows with the group, to 3.26 and 3.64 for eleven windows, while the fidelity falls to 0.62 and 0.57: a further window costs REMAKE only its maps, 3% of a record for $K = 4$ and 8% to 10% for $K = 11$ on eTraM, and costs a subset scheme every event that it keeps. On DSEC $G = 1.31$ (1.29 to 1.34) for four windows and 2.71 (2.63 to 2.88) for eleven, and 1.32 for four windows over the 31 sequences from Zurich. The key events that REMAKE sends are 21% of the events of a group of four windows by day, 17% by night and 22% on DSEC, and 7%, 6% and 8% of a group of eleven. The receiver rebuilds the rest. Fig. 4 shows the rebuilt events of a typical group of each stratum.

Short groups. For $K = 2$ REMAKE loses, with $G = 0.92$ by day, 0.94 by night and 0.81 on DSEC. Half of such a group is its key window, and a key event costs more than an event of a subset: a map moves it by whole pixels, so it is coded at one pixel where the subset is coded at the two pixels of the grid. The maps repay this in groups of four windows and more.

Motion. Key events that are repeated in place, without maps, stay below the subset schemes for $K = 4$, with $G_{\text{rep}} = 0.60$, 0.90 and 0.88. The gain of REMAKE is therefore the gain of its maps, a factor 2.57, 1.82 and 1.48 over repetition (the median of the ratio per recording). Repetition does better where true events stay in the regions. Ten windows after the key window, 25% of the true events of eTraM by day still fall into the regions, and 70% on DSEC, where the whole scene moves. A repeated event there meets some other event in its voxel, and the fidelity of repeated windows falls only from 0.39 to 0.26 over ten windows on DSEC, against 0.18 to 0.07 on eTraM by day. Such pairs do not need the right mo-



Figure 4: Signed event counts per pixel, positive minus negative polarity, of a typical group: the key events that REMAKE sends, the true events of window $m = 4$, and the events that the receiver rebuilds for it from the stream.

Table 1: REMAKE on the working grid, matched coder. b and b_{sub} in bits per event, G_{rep} for the repeated key events, with the ratio taken per recording, G_{T} with the transport coder.

	K	F	b	b_{sub}	G	G/G_{rep}	G_{T}
eTraM, day	4	0.71	0.42	0.65	1.55	2.57	2.68
	11	0.62	0.16	0.53	3.26	4.13	5.96
eTraM, night	4	0.65	0.35	0.57	1.62	1.82	2.85
	11	0.57	0.13	0.48	3.64	2.45	6.62
DSEC	4	0.67	0.52	0.66	1.31	1.48	2.38
	11	0.58	0.20	0.55	2.71	1.59	5.20

tion. For eleven windows repetition exceeds the subset schemes at night and on DSEC (1.40 and 1.63), and the ratio G/G_{rep} removes this part.

Threshold. From $\psi = 1/16$ to $\psi = 1$, the bits for $K = 4$ fall from 0.62 to 0.42 by day, from 0.74 to 0.35 by night and from 0.72 to 0.52 on DSEC, the fidelity falls from 0.75 to 0.71, from 0.74 to 0.65 and from 0.71 to 0.67, and G rises from 1.17 to 1.55, from 0.93 to 1.62 and from 1.06 to 1.31. The blocks that a lower threshold adds cost key events and return few pairs in the later windows. At $\psi = 1$ the regions are 1% of the blocks of eTraM and hold 83% of the events of the key window by day and 69% by night. On DSEC they are 31% of the blocks and hold 86%.

Affine map against translation. With one translation per block in place of the affine map, the fidelity for $K = 11$ is lower by 0.02 by day, 0.01 by night and 0.03 on DSEC. In gain the translation is ahead on eTraM, by 1% to 3%, because the shape parameters cost more bits there than their pairs return, and the affine map is ahead on DSEC, by 2.5% for $K = 4$. On DSEC the whole scene moves with the camera: ten windows after the key window the median block has moved by 38 pixels, and 91% of the blocks have a nonzero shape parameter.

Coders. With the transport coder all events cost 23.4, 24.0 and 22.0 bits each, and with the matched coder 1.56, 2.04 and 1.64. A receiver that counts on the working grid needs 7% to 9% of the bits of the transport. The rest buys precision that this receiver does not use. Against the transport coder REMAKE gains more, 2.4 to 2.9 for four windows and 5.2 to 6.6 for eleven, because that coder leaves more to save. We take the matched coder as the reference.

Lossless reference. A fidelity of one on a voxel grid is not a lossless reconstruction. For the lossless cost, we computed the ideal code length of the events inside annotated vehicles of eTraM that are isolated and move at 20 to 300 pixels per second, with addresses of one pixel and times of one microsecond. An event costs 18.3 bits under a density that is uniform over the box of the vehicle and the window. A density that is built from the events of the preceding 100 ms saves 0.5 bit by day and 1.2 by night when these events are left in place, and 2.9 and 3.0 bits when they are moved along the estimated

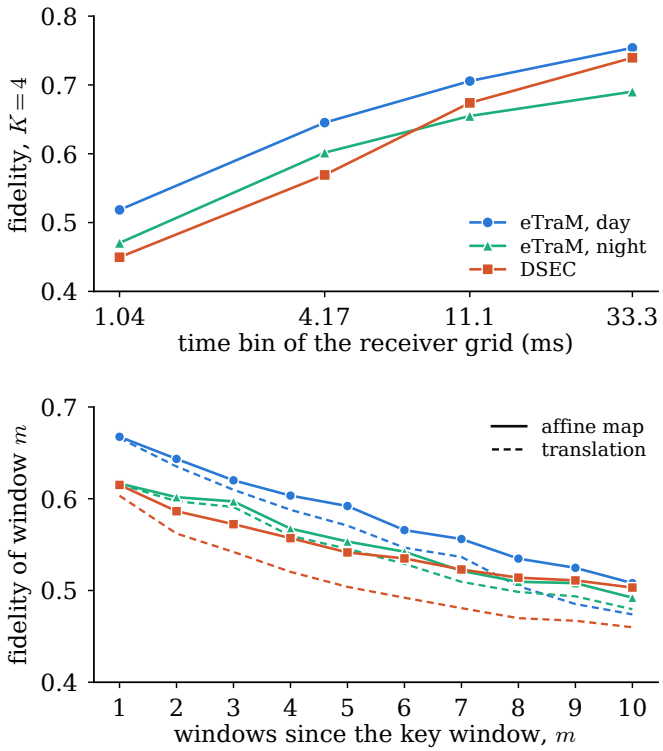


Figure 5: Top: fidelity of a group of four windows against the time bin of the receiver grid, at $\Delta_x = 2$ pixels. Bottom: fidelity of window m on the working grid, for the affine map (solid) and the translation (dashed).

motion. The motion thus accounts for about a sixth of the lossless cost. The remainder is private to the pixels, as Theorem 1 states for the model.

6.4 Where Regeneration Stops Working

Tolerance. The time bin of the receiver grid plays the role of the tolerance. From a bin of 33.3 ms to one of 1.04 ms, the fidelity of a group of four windows falls from 0.75 to 0.52 on eTraM by day, from 0.69 to 0.47 by night and from 0.74 to 0.45 on DSEC (Fig. 5, top). The transport coder serves every grid, and with it the gain falls from 2.95, 3.07 and 2.73 to 1.69, 1.81 and 1.35 over this range, and to 1.24, 1.42 and 1.01 on the grid of one pixel and 1.04 ms. Moved events carry the phases and the jitter of the pixels they came from (Proposition 2), and by (33) a train whose offset to the true one is uniform cannot fill bins shorter than the event spacing. This is the scaling of the threshold of Theorem 2. The event spacings of the recorded pixels were not measured, so the comparison is one of trends. The same trend separates objects. With annotated vehicles as regions and one translation per vehicle, the fidelity for $K = 4$ on the working grid rises from 0.52 below 20 pixels per second to 0.72 above 100 by day, and from 0.61 to between 0.71 and 0.74 by night: by (8) a faster edge has a shorter event spacing, so the same bin is a larger tolerance for it.

Distance from the key window. The fidelity of window m falls with m, from 0.67 to 0.51 on eTraM by day, from 0.62 to 0.49 by night and from 0.62 to 0.50 on DSEC over ten windows (Fig. 5, bottom). The number of reconstructed events does not cause the fall: its ratio to the number of true events moves from 0.83 to 0.80 by day and from 0.86 to 0.83 on DSEC. The moved events fall into the wrong voxels. The translation loses more than the affine map, and the difference grows with m, to 0.03 by day, 0.01 by night and 0.04 on DSEC at the tenth window, as the residual of a block translation against the flow accumulates. Two losses are present from the first window on, where the fidelity is 0.62 to 0.67: the events outside the regions, which hold the key window itself to 0.91, 0.82 and 0.92, and the phases and jitter that moved events carry.

A typical group. Fig. 4 shows both losses in one group of each stratum. The group is taken from the recording whose fidelity of window $m = 4$, formed from the sums over its groups, is the median of the stratum, and it is the group of that recording whose fidelity is closest to the recording's. Its fidelity is 0.61 by day, 0.57 by night and 0.56 on DSEC. A first version of the figure took the median group, which weights every group equally and by day picked a sparse group of fidelity 0.21. The rule above aggregates as the tables do. Where a region holds the key events, the rebuilt edges lie where the true ones are. Where none does, nothing is rebuilt, as for the road in the lower half of the DSEC view, whose blocks hold fewer than ψB^2 events in the key window. Inside the regions the rebuilt edges are dotted where the true ones are filled, since a map moves every key event to one pixel, and a map that enlarges the region spreads the same events over more pixels.

7 Conclusion

The pixels on the path of a moving edge share the edge and its motion, and each has a threshold phase and a jitter of its own. A lossless code pays for the phase and the jitter at every pixel. A timing tolerance removes first the jitter and then the phase from what a code must describe, and beyond a threshold set by the event spacing the events are regenerated from the edge and the motion alone. REMAKE, a codec built on this, moves key events along affine maps on a block grid. Calibrated on a few recordings of each dataset, it needs fewer bits than schemes that send a subset of the events at equal fidelity once a group holds four windows, on a static camera and on a moving one. Its advantage shrinks as the time bin of the receiver shortens, in line with the threshold of the theory. Two steps remain open. REMAKE moves a fixed number of events, keeps its regions for a whole group and its group length and threshold for a whole stream, where a change of scale changes the count, new objects enter, and the best group length and threshold vary with the

scene. And the theory covers one ramp edge in uniform passage, so texture, noise events and threshold mismatch are still to be modeled.

A Proofs of Lemmas 1 and 2

Proof of Lemma 1. The reference starts in $a_i + C\mathbb{Z}$ by the definition of the phase. By (A1) it changes by $\pm C$, so it stays in this set, and an event requires $\ell_i = \mu_i \pm C$. □

Proof of Lemma 2. By (A2) and (A2′), ℓ_i is nondecreasing, equal to 0 before t_i, to $\gamma r_i(t - t_i)$ during the passage, and to V after it. Since $\mu_i \le 0 = \ell_i$ before the edge arrives and $\mu_i = a_i - C$, the events occur when ℓ_i reaches the levels $a_i + (k-1)C$, $k = 1, 2, \dots$, as long as the level is at most $V = nC$. Because $0 < a_i \le C$, exactly the levels $k = 1, \dots, n$ are reached. Solving $\gamma r_i(t - t_i) = a_i + (k-1)C$ gives (10), and a_i uniform on $(0, C]$ makes A_i uniform on $(0, \tau_i]$. □

B Proof of Theorem 1

By Lemma 2, $T_i = c_i + A_i \mathbf{1} + \varepsilon_i$ with c_i a function of the template and the flow, $\mathbf{1}$ the all-ones vector, and $\varepsilon_i \sim \mathcal{N}(0, \sigma^2 I_n)$. Independence over i follows from (A3) and (A4). Let Q be an orthogonal matrix whose first row is $\mathbf{1}^\top/\sqrt{n}$. Then $Q(T_i - c_i) = (X_i, W_i)$ with

$$X_i = \sqrt{n}\, A_i + \varepsilon_i^{\parallel}, \qquad W_i \sim \mathcal{N}(0, \sigma^2 I_{n-1}), \tag{34}$$

where $\varepsilon_i^{\parallel} \sim \mathcal{N}(0, \sigma^2)$. The variables X_i and W_i are independent, because $Q\varepsilon_i$ has independent coordinates and A_i is independent of ε_i. Differential entropy is invariant under translations and orthogonal maps, so $h(T_i) = h(X_i) + h(W_i)$, which is (11).

Let f be the density of T_i and f_δ its average over the cells of the clock lattice. A cell has probability $\delta^n f_\delta$, so $H(Y_i \mid g, \phi) = n \ln(1/\delta) - \int f \ln f_\delta$ and $e_\delta = \int f \ln(f/f_\delta)$ is a relative entropy, which is nonnegative. The density f is a mixture over A_i of $\mathcal{N}(m, \sigma^2 I_n)$ densities and f_δ is the same mixture of their cell averages. By the joint convexity of relative entropy, e_δ is at most its largest value for a single $\mathcal{N}(m, \sigma^2 I_n)$ density, which is n times the value for a scalar normal density and depends only on δ/σ and on the offset of m in a cell. It tends to zero with δ/σ, uniformly in the offset [24, Ch. 8]. ■

C Proof of Theorem 2

The distortion (14) is invariant under the translation by c_i and the orthogonal map Q, so R_i is the rate-distortion function of the independent components (X_i, W_i) of (34) under the total squared error nD. Their variances are s_i^2 and σ^2, the latter $n - 1$ times. We drop the index i and write η for $\eta(\sqrt{n}\,\tau_i, \sigma)$ and $\ln^+ x = [\ln x]^+$.

Lower bound (17) *and* (18). Let $(\hat{X}, \hat{W})$ have component errors d_X and $d_1, \dots, d_{n-1}$ with $d_X + \sum_j d_j \le nD$. Independence of the components gives $I(X, W; \hat{X}, \hat{W}) \ge I(X; \hat{X}) + \sum_j I(W_j; \hat{W}_j)$. For a Gaussian component, $I(W_j; \hat{W}_j) \ge \frac{1}{2} \ln^+(\sigma^2/d_j)$, and $I(X; \hat{X}) \ge h(X) - h(X - \hat{X}) \ge \eta - \frac{1}{2} \ln(2\pi e\, d_X)$. The function $\ln^+(\sigma^2/d)$ is convex in d, so the Gaussian terms are smallest when all d_j equal a common d_W, and $d_W > \sigma^2$ brings no benefit. For $n \ge 2$ this gives

$$R(D) \;\ge\; \min_{d_W} \Big[\eta - \tfrac{1}{2} \ln\big(2\pi e\,(nD - (n-1)d_W)\big) + \tfrac{n-1}{2} \ln \frac{\sigma^2}{d_W} \Big], \tag{35}$$

with the minimum over $0 < d_W \le \sigma^2$, $d_W < nD/(n-1)$. The bracket is convex in d_W and its derivative vanishes only at $d_W = D$. The minimum is at $d_W = D$ when $D \le \sigma^2$, where it equals the right side of (17), and at $d_W = \sigma^2$ otherwise, where it equals $U(D) - \Gamma$. For $n = 1$ there is no W and the same values follow with $d_X = D$.

Equality for $D \le \sigma^2$. Write $\varepsilon = \varepsilon' + Z$ with independent $\varepsilon' \sim \mathcal{N}(0, (\sigma^2 - D) I_n)$ and $Z \sim \mathcal{N}(0, D I_n)$, and let $\hat{T} = c + A\mathbf{1} + \varepsilon'$. Then $T = \hat{T} + Z$ with Z independent of $\hat{T}$, the distortion is D, and $I(T; \hat{T}) = h(T) - h(Z)$, which is the lower bound.

Upper bound for $\sigma^2 < D < D_A$. Spend no rate on W, at error σ^2 per component, and describe X at error $d = nD - (n-1)\sigma^2 \in (\sigma^2, s^2)$ through $\hat{X} = (1 - d/s^2)(X - \mathbb{E}X + Z') + \mathbb{E}X$, with Z' an independent normal variable of variance $s^2 d/(s^2 - d)$. The error is d, and $I(X; \hat{X}) = I(X; X + Z') \le \frac{1}{2} \ln(s^2/d)$, because a normal input maximizes the mutual information across an additive normal noise channel at a given input variance [24].

Lower bound (21) *and zero rate.* For a variable Ξ with $\ln \mathbb{E}\, e^{\lambda(\Xi - \mathbb{E}\Xi)} \le \lambda^2 \varsigma^2/2$ for all λ under a law P, and any law Q, the nonnegativity of the relative entropy between Q and the law with density proportional to $e^{\lambda \Xi}$ with respect to P gives $D(Q\|P) \ge \lambda(\mathbb{E}_Q \Xi - \mathbb{E}_P \Xi) - \lambda^2 \varsigma^2/2$, and the best λ gives $(\mathbb{E}_Q \Xi - \mathbb{E}_P \Xi)^2 \le 2\varsigma^2 D(Q\|P)$. With P the law of X and Q its conditional law given $\hat{T}$, averaging over $\hat{T}$ yields $\mathrm{Var}\, \mathbb{E}[X \mid \hat{T}] \le 2\varsigma^2 I(X; \hat{T})$. The uniform part of X has width $a = \sqrt{n}\,\tau$ and satisfies the condition with $a^2/12$, since $\ln(\sinh x/x) \le x^2/6$, so X satisfies it with $\varsigma^2 = s^2$, and each W_j with $\sigma^2 \le s^2$. Since the conditional mean minimizes the squared error and the components are independent,

$$\begin{aligned} nD &\ge \mathbb{E}\|T - \mathbb{E}[T \mid \hat{T}]\|^2 \\ &= nD_A - \mathrm{Var}\, \mathbb{E}[X \mid \hat{T}] - \sum_j \mathrm{Var}\, \mathbb{E}[W_j \mid \hat{T}] \\ &\ge nD_A - 2s^2 I(T; \hat{T}), \end{aligned} \tag{36}$$

which is (21). With no description the best output is the mean, which is (22), at distortion $\frac{1}{n}(s^2 + (n-1)\sigma^2) = D_A$, so (19) holds. ■

D Proof of Theorem 3

The vectors T_i are independent and the distortion is additive over i. For any $\hat{T}$ with average distortion at most D, let $D_i = \mathbb{E}\, d(T_i, \hat{T}_i)$. Then $I\big((T_i)_i; (\hat{T}_i)_i\big) \geq \sum_i I(T_i; \hat{T}_i) \geq \sum_i R_i(D_i)$ with $\frac{1}{M}\sum_i D_i \leq D$. Conversely, independent test channels whose rates are within any given margin of $R_i(D_i)$ come within M times that margin of the sum. Each R_i is convex, nonincreasing and continuous, and tends to infinity as $D_i \to 0$ by (17), so the minimum in (23) exists.

For $D < \sigma^2$, every R_i is differentiable at D with the derivative $-n/(2D)$ by (17), and it is convex on $(0, \infty)$. Hence $R_i(D_i) \geq R_i(D) - \frac{n}{2D}(D_i - D)$ for every $D_i > 0$, and summing over i with $\sum_i (D_i - D) \leq 0$ shows that the allocation $D_i = D$ is optimal. This gives (24) for $D < \sigma^2$, and the case $D = \sigma^2$ follows by continuity.

By Theorem 2, $\sum_i R_i(D_i) = 0$ if and only if $D_i \geq D_{A,i}$ for every i, and such an allocation with average at most D exists if and only if $D \geq \frac{1}{M}\sum_i D_{A,i} = \bar{D}_A$. Finally, (21) and $s_i^2 \leq n\tau_{\max}^2/12 + \sigma^2$ give $\sum_i R_i(D_i) \geq n \sum_i [D_{A,i} - D_i]^+ / (2(n\tau_{\max}^2/12 + \sigma^2))$, and $\sum_i [D_{A,i} - D_i]^+ \geq M(\bar{D}_A - D)$, which is (26). ■

E Proof of Proposition 1

Divide side j of the box into the smallest number of equal cells of length at most $\varsigma_j = \sqrt{\xi}\,\sigma/(q\, S_j M^{\beta_j})$, and let $\hat{\theta}$ be the center of the cell that contains θ. Side j has at most $L_j M^{\alpha_j}/\varsigma_j + 1$ cells, so the index of $\hat{\theta}$ takes $\sum_j \ln\big(L_j S_j q\, M^{\alpha_j + \beta_j}/(\sqrt{\xi}\,\sigma) + 1\big)$ nats, which is (27). The decoder outputs $\hat{T}_{i,k} = \bar{t}_{i,k}(\hat{\theta})$. The segment from θ to $\hat{\theta}$ lies in the box, so by the mean value theorem $e_{i,k} = \bar{t}_{i,k}(\theta) - \bar{t}_{i,k}(\hat{\theta})$ satisfies $|e_{i,k}| \leq \sum_j S_j M^{\beta_j} \varsigma_j / 2 \leq \sqrt{\xi}\,\sigma/2$. By (10), $T_{i,k} - \bar{t}_{i,k}(\theta) = A_i - \tau_i/2 + \varepsilon_{i,k}$ has mean zero given θ and variance $D_{A,i}$, and $e_{i,k}$ is a function of θ. Hence $\mathbb{E}\,(T_{i,k} - \hat{T}_{i,k})^2 = D_{A,i} + \mathbb{E}\, e_{i,k}^2 \leq D_{A,i} + \xi\sigma^2/4$, and the average over the path is at most $\bar{D}_A + \xi\sigma^2/4 \leq (1+\xi)\bar{D}_A$. ■

F Proof of Proposition 2

By (10) and (A4), $T_{i,k} - \tilde{T}_{i,k} = (A_i - A_j) + (k-1)(\tau_i - \tau_j) + \varepsilon_{i,k} - \varepsilon_{j,k}$. The phases and the jitter of the two pixels are independent, so this difference has the mean $(k - \frac{1}{2})(\tau_i - \tau_j)$ and the variance $\tau_i^2/12 + \tau_j^2/12 + 2\sigma^2 = D_{A,i} + D_{A,j}$. Its mean square is the variance plus the squared mean, and $\frac{1}{n}\sum_{k=1}^{n}(k - \frac{1}{2})^2 = (4n^2 - 1)/12$. ■

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